

DATASCI 406 FA 26, Quiz 2.1

Name	Uniqname	UMID

Each question is worth 1 point. Circle one answer for each.

Question 1 (1 point)

For a random variable X and a probability p , what does the quantile function $Q_X(p)$ return?

1. The probability that X is less than or equal to p .
2. The smallest value x such that the CDF evaluated at x is at least p .
3. The value x at which the density of X equals p .
4. The value of the CDF evaluated at X .

Solution

Answer: 2. This is $Q_X(p) = \inf\{x : F_X(x) \geq p\}$ read aloud. Option 1 is the CDF evaluated at p , and option 3 confuses the density with the CDF. Option 4 is the probability integral transform $F_X(X)$, which is a $U(0, 1)$ random variable rather than a value of x .

Question 2 (1 point)

X has CDF $F(x) = x^3$ for $0 \leq x \leq 1$. What is its quantile function $Q(u)$ for $0 \leq u \leq 1$?

1. u^3
2. $3u^2$
3. $1 - u^{1/3}$
4. $u^{1/3}$

Solution

Answer: 4. F is continuous and strictly increasing, so solve $u = x^3$ for x . Option 1 is F again, and option 2 is the density $f(x) = 3x^2$.

Question 3 (1 point)

Which line of R produces 1000 draws from an exponential distribution with rate 2, using `qexp()` and not `rexp()`?

1. `runif(1000, qexp(0, rate = 2), qexp(1, rate = 2))`
2. `qexp(1000, rate = 2) * runif(1000)`
3. `qexp(runif(1000), rate = 2)`
4. `runif(qexp(1000, rate = 2))`

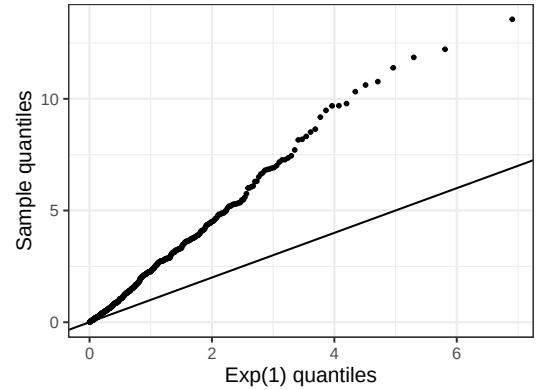
Solution

Answer: 3. Draw $U \sim U(0, 1)$, then apply Q to it. Option 1 is uniform over the support (which here is unbounded, so it fails outright). Options 2 and 4 pass 1000 to `qexp()` as though it were a probability.

Question 4 (1 point)

A sampler is meant to draw from $\text{Exp}(1)$. The QQ plot at right compares 500 of its draws with the $\text{Exp}(1)$ quantiles, and the line is the 45 degree line. Is the sampler correct?

1. Correct: the points form a straight line.
2. Wrong: the points do not follow the 45 degree line.
3. Can't tell: a plot is not a hypothesis test.



Solution

Answer: 2. The points follow a line of slope about 2, not the 45 degree line. (Reading the direction, each sample quantile is about twice the matching $\text{Exp}(1)$ quantile, but the quiz does not ask for that.) The sampler used $-2 * \log(\text{runif}(n))$, which is $\text{Exp}(1/2)$. A straight line means the right family at the wrong scale; only the 45 degree line means the right distribution.

Question 5 (1 point)

X takes the values below. To draw X you generate $u = 0.53$ and apply the inverse transform method for discrete variables. What value x do you generate?

x	10	20	30	40
$\Pr(X = x)$	0.2	0.3	0.1	0.4

1. 10 2. 20 3. 30 4. 40

Solution

Answer: 3. The running totals are 0.2, 0.5, 0.6, 1.0. The total first reaches 0.53 at $x = 30$, since $F(20) = 0.5 < 0.53 \leq 0.6 = F(30)$. Option 2 stops at the last total below u instead of the first one above it. Option 4 picks the largest probability, or the one closest to u , which skips the running total entirely.

Rubric. 1 pt per question, no partial credit.